

Pavan K M

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SUMMARY

Full-stack software engineer with 3+ years of experience building and shipping production systems — from scalable backend APIs and React applications to AI-powered agents and cloud infrastructure. Founded and led engineering at KeyChainN as CTO, taking a travel loyalty platform from zero to live users. Hands-on experience across the entire stack: REST APIs, microservices, LLM agents, RAG pipelines, payment integrations, and AWS deployments. Proven ability to own products end-to-end, lead small engineering teams, and ship fast without sacrificing quality.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML/CSS

Frontend and Frameworks: React.js, Django, Flask, FastAPI, Spring Boot, component-based UI, merchant dash-boards

APIs and Architecture: REST APIs, GraphQL, microservices, async services, client-server architecture, distributed systems

AI and ML: RAG pipelines, embeddings, vector similarity search, ranking systems, LangChain

Databases: PostgreSQL, Oracle, MongoDB, Redis, OpenSearch, DynamoDB, RDS, MySQL

Data Platforms: Apache Spark, Apache Kafka, Celery, event-driven pipelines, batch processing, Great Expectations, Cerberus, marshmallow

AWS and Cloud: AWS (EC2, ECS, Lambda, S3, RDS, API Gateway, CloudWatch), Google Cloud Platform, Docker, Kubernetes, GitHub Actions

Tools: Git, Cursor, Claude Code, Terraform, Notion, Postman, PyTest, JWT, OAuth2, RBAC

Practices: CI/CD, Jenkins, monitoring, incident re-sponse, Agile, Scrum

EXPERIENCE

KeyChainN - keychainn.com

Chief Technology Officer

Aug 2025 – Present

- Architected and launched a full-stack travel loyalty platform from zero, owning every layer: backend services, frontend, cloud infrastructure, and third-party integrations with real users live in production
- Personally designed and built the entire transaction stack from booking flows and availability management to payment integrations and audit infrastructure without delegating the core engineering to anyone else
- Drove product definition end-to-end: authored PRDs, led system design sessions, and translated roadmap into executable engineering work tracked across GitHub and Notion
- Hired, onboarded, and led a team of 3-4 engineers running code reviews, setting technical standards, and maintaining velocity without sacrificing quality
- Designed and shipped partner onboarding infrastructure, enabling property managers to integrate into the network with minimal friction
- Represented the company at Startup Grind (Startup Mania), pitching the technical vision and product architecture to investors and ecosystem partners

Stack: React, Node.js, Vite, AWS, Git, Cursor

Evergreen

Software Engineer

Remote

Oct 2021 – Jun 2023

- Took full ownership of a greenfield project and delivered a full-stack cloud infrastructure cost comparison platform end-to-end in a single quarter compressing deployment time from 2 weeks to 2 days
- Built and integrated an AI-powered pricing recommendation engine on top of live AWS pricing APIs, making infrastructure cost modeling dynamic and real-time
- Enabled engineering managers to browse prebuilt Terraform templates, estimate costs, and deploy directly, eliminating manual DevOps overhead for new project setup

Stack: React, Node.js, AWS, Terraform — built with Cursor and Claude Code

Syniti

Software Engineer

Remote

Jan 2020 – Aug 2021

- Built and maintained a full-stack data validation and consistency web application using Python, Flask, and React, enabling enterprise clients to define, execute, and monitor multi-layered data quality rules across heterogeneous data sources in a unified dashboard.
- Implemented a configurable data validation engine using Great Expectations and Cerberus, enforcing schema contracts, null constraints, referential integrity, and cross-field consistency checks across structured datasets ingested from PostgreSQL, CSV, and REST API sources.
- Designed Python-based data consistency pipelines using marshmallow for serialization and schema validation, reconciling records across upstream and downstream systems and detecting duplicates, value drift, and format violations at ingestion boundaries.
- Surfaced data quality discrepancies through a React dashboard with drill-down filtering and exportable audit reports, giving enterprise clients real-time visibility into validation failures, anomaly trends, and dataset health across environments.

- Built background validation job orchestration using Celery and Redis, decoupling long-running data quality checks from synchronous Flask API responses and enabling scheduled and on-demand validation runs across large data volumes without blocking client-facing workflows.
- Developed Flask REST API endpoints exposing validation rule management, job status tracking, and quality metrics to the frontend, with role-based access control via JWT enforcing client and administrator permission tiers across all API surfaces.
- Integrated PostgreSQL as the primary metadata and results store, designing normalized schemas for validation rule definitions, execution history, anomaly logs, and dataset registrations; optimized query performance through indexing and materialized views for reporting workloads.
- Containerized the full application stack with Docker and deployed on AWS EC2 with S3 used for bulk dataset uploads and validation report archival; implemented CloudWatch alerting for job failures and pipeline SLA breaches across client environments.
- Wrote comprehensive PyTest test suites covering validation logic, Flask API contract correctness, and edge case data scenarios, maintaining high coverage across the validation engine and supporting confident iterative releases.

Nerds & Geeks

Software Engineer

Remote

Jan 2020 – Aug 2021

- Built a native mobile and web application for an international e-commerce client, reaching 100K+ users and driving a 35% increase in sales post-launch
- Developed secure RESTful APIs using Spring MVC with JPA and Hibernate, improving data access speed by 20% and backend reliability by 15%
- Integrated Stripe for automated billing and Twilio for notifications, cutting manual ticket resolution steps by 50% and improving alert delivery latency by 60%
- Designed async service-to-service APIs for distributed microservices, improving cross-module communication efficiency by 40%

EDUCATION

University of Texas at Dallas

Master of Science in Information Technology

Jawaharlal Nehru Technological University

Bachelor's in Computer Science and Engineering

Projects

ApplyFlow AI — applyflowai.com

Built in 2 weeks · 2025

- Built a full-stack AI job application agent solo in 2 weeks — live in production with real users
- Engineered an autonomous bot that discovers relevant job listings and auto-applies based on user-defined preferences, eliminating hours of manual application overhead
- Built an AI-powered recruiter outreach system that drafts personalized messages and sends them automatically — not generic blasts
- Designed a context-based learning engine: when the bot encounters an unknown application field, it uses the user's resume and the JD to infer the answer, then stores it to get smarter over time
- Shipped a React dashboard for users to track their full application pipeline, review submissions, and manage resume variants

Stack: Python, TypeScript, React — built with Claude Code